

Genesis_Medicine: an open-source AI pipeline for Korean traditional medicine drug discovery — ~25 active modules + ~25 adapter scaffold catalog with design philosophy (v0.3 audit-corrected)

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Abstract

Modern AI-driven drug discovery is built on a heterogeneous stack of open-source tools. Specialized application domains require tool selection, integration, and adaptation. We describe **Genesis_Medicine**, an open-source pipeline for **AI-driven Korean traditional medicine (한약·생약) drug discovery**. We honestly distinguish: (i) a **7-tool active core** that produces all real pipeline outputs in this work (Boltz-2 cofold, REINVENT 4 generative, ADMET-AI v2 property prediction, OpenMM 8 + openmmtools alchemical sampling, RDKit chemistry, ChEMBL read-only, Open Targets v4 GraphQL); and (ii) a **40+ adapter scaffold catalog** organized into 12 functional tiers, written as Python modules and exposed through a natural-language agent interface but **not yet actively used in primary pipeline runs** — including AlphaFold-3-class cofold ensemble (Chai-1, Protenix-v2, OpenFold-3), generative SOTA (PocketXMol, FlowMol3, DiffSBDD, TurboHopp), ML potentials (MACE-OFF24, AIMNet2), pose validation (PoseBusters), and specialized adapters (PROTAC designer, chronotherapy, TxGNN repurposing). The active core supports the complete workflow from natural-product curation through corrected ABFE; the adapter scaffold provides ready integration points for prioritized future expansion. We discuss the design philosophy — 3-pillar institutional integration (HAN PREDICT, Inc. AI healthcare platform; Recover Korean Medicine Clinic), commitment to honest in silico reporting, and explicit method limitations including MMP-1 zinc handling and Boltz-2 binary-classifier vs IC₅₀ distinction. The pipeline is open-source under Apache-2.0 at https://github.com/crazat/genesis_medicine.

Keywords: open-source pipeline, Korean traditional medicine, AI drug discovery, REINVENT4, Boltz-2, ADMET-AI, ABFE, OpenMM, natural products, integrated stack.

Plain-language summary

Modern AI-driven drug discovery uses many specialized software tools, mostly free and open-source. Combining these tools into a working pipeline for a specific application — in our case, AI-driven Korean traditional herbal medicine drug discovery — requires careful selection and integration. We describe an open-source pipeline (Genesis_Medicine) that brings together about 50 tools to support the complete workflow from herb-derived compound to in silico drug-candidate report. The code is freely available for academic and commercial use. **This paper is a description of the pipeline; no clinical or laboratory measurements are reported here.**

1. Introduction

1.1 The fragmented AI drug-discovery stack

Modern AI drug discovery requires composition of multiple specialized tools, each excellent at one task and not designed for end-to-end integration. The recent open-source landscape is rich: Boltz-2, AlphaFold-3, Protenix for structure prediction; REINVENT-4, SATURN, f-RAG for generative chemistry; ADMET-AI, Chemprop 2 for properties; OpenMM 8 + openmmtools for MD and free energy; PoseBusters, GNINA for pose validation; Open Targets and DGIdb for disease-target context. Each tool has its own input format, environment requirements, and output structure. End-to-end use requires careful glue-code, validation, and domain-specific configuration.

1.2 The Korean traditional-medicine domain

The Korean traditional-medicine (한약·생약) domain has specific requirements that are not fully addressed by general-purpose pipelines:

- **Natural-product chemistry** (saponins, terpenoids, anthraquinones, polyphenols, alkaloids) — atypical scaffolds for which Caucasian-cohort-trained models may underperform.
- **Korean Pharmacopoeia (KP) and Korean Herbal Pharmacopoeia (KHP)** — specific listings, quality standards, traditional indications.
- **다중성분 복합처방 (multi-component formulations)** — Korean medicine prescribes combinations, not single compounds; pipeline must support combination scoring.
- **체질의학 / 사상의학 (constitutional medicine)** — patient stratification by traditional constitutional theory.
- **Regulatory: MFDS, 의료법 §56, 화장품법** — Korean-specific compliance.

1.3 Genesis_Medicine: design goals

We assembled the Genesis_Medicine pipeline with the following goals:

1. **End-to-end integration** of generative → property → structure → MD → ABFE → manuscript.
 2. **Korean traditional-medicine domain orientation:** KP / KHP herb-compound mapping, 한약 처방 network analysis, Korean PGx panel.
 3. **Honest in silico reporting:** TRIPOD-AI compliance, explicit limitations, retraction of incorrect prior results when discovered.
 4. **Open-source under permissive license:** Apache-2.0; commercial-use compatible; avoids GPL viral.
 5. **Natural-language interface:** a single LLM-agent that accepts natural-language requests and dispatches to the appropriate pipeline tier.
 6. **3-pillar institutional integration:** with HAN PREDICT (AI healthcare platform) and Recover Korean Medicine Clinic; clinical feedback loop closed.
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2. Pipeline architecture: active core + adapter catalog

2.0 Active modules (~25, with real pipeline outputs)

A v0.3 internal audit (2026-04-26) revised our earlier framing. The active set is broader than initially reported:

Core compute: Boltz-2 v0.6.1 (MIT) — cofold + affinity (240+ cofolds across rounds + 4 disease screens); **REINVENT 4** v4.4 (Apache-2.0) — mol2mol scaffold-hopping; **ADMET-AI** v2.0.1 (MIT) — 6 endpoints; **OpenMM 8.5.1** + **openmmtools 0.26** (MIT) — 10 ns MD + 16-window alchemical replica exchange ABFE; **RDKit** (BSD-3); **PoseBusters** 0.6.5 (queued for cofold pose validation); **Chai-1r** (Apache-2.0, 2025-Q4 fully open) — installed for 2-way ensemble; **GFN2-xTB** via xtb-python 22.1 — semi-empirical QM single-point (EMB-3 -50.09 Hartree, Embelin -65.90 Hartree computed in this work).

Database integration: ChEMBL v34 (CC-BY-SA), Open Targets v4 GraphQL (CC0; 11 disease IDs + 15 target IDs production-cached pipeline).

Workflow + reproducibility: **Prefect 3** DAG framework (11-stage pipeline orchestration); **Docker** + docker-compose containerization; **pyproject.toml** with ruff + mypy strict-mode; pinned dependencies; **GitHub Actions CI/CD** (lint + test + auto-PDF-build for preprints); 16 test files / 95 unit tests with GPU/slow/network markers.

Domain-specific active modules: residence_time.py (Eyring + Jarzynski + steered MD); spatial_atlas.py (Visium CytAssist + Xenium IPF atlas); scfoundation_adapter.py (with linear baseline check); hira_qaly.py (Korean health economics); stratum_corneum.py (ceramide/FFA/cholesterol bilayer builder); benchmark_baseline.py (ROC-AUC + Mann-Whitney U); logKp head (FDA 2326 LGBM, ≥-2 topical gate); chronotherapy/jaoryuju.py (12-meridian + circadian); donguibogam_miner.py (Korean classical text mining); fpocket cryptic_scan (TGFB1/MMP1/CTGF allosteric scan).

Recently activated (2026-04-26): BioEmu 1.3.1 (Microsoft 2026 protein equilibrium ensembles); deeptime 0.4.5 + PyEMMA 2.5.12 (MSM-based binding kinetics); xtb-python 22.1 (GFN2-xTB QM correction); 5 missing-target MSAs (MITF, SRD5A1, SREBP1, FBN1, mTOR); production Open Targets agent dispatch.

Outputs: all .csv and .json files in pilot/ are products of this active module set.

2.1 Adapter scaffold catalog (40+ modules)

Tier 0-11 below describe Python modules registered in `agents/all_tiers_tools.py` exposing 70+ natural-language tool dispatches. **These are integration scaffolds**, mostly subprocess/Python wrappers over external open-source tools that have not yet been run in the present pipeline. They are documented here for reproducibility of the integration architecture, not as active capability claims.

Tier 0 — Foundation (~7 modules)

Open-source SOTA stack: Boltz-2 cofold, REINVENT 4 generative, ADMET-AI predictions, RDKit chemistry, OpenMM 8 MD, openmmtools alchemical sampling, MACE-OFF24 ML potential. Standard conda / pip / uv installation under Korean GPU stack (CUDA 12.8, RTX 5090 Blackwell).

Tier 1 — Acceleration / Quality (~6 modules)

Protenix-v2, OpenFold-3 ensemble (Boltz-2 consensus); AlphaFlow-Lit; CarsiDock-Cov for covalent-warhead screening; Boltz-ABFE direct path; AIMNet2 ML potential; DeepRetro retrosynthesis.

Tier 2 — Skin-specific Adapters (~6 modules)

PanDerm (skin lesion vision-language; integrated as skin-classification adapter); ChemCrow LLM agent wrapper; stratum-corneum MD setup; Bayesian SAR optimization; ESM3 protein function; scFoundation single-cell.

Tier 3 — Multi-target / Cryptic-site (~5 modules)

Four added skin-fibrosis targets (LPA1 / $\alpha\text{v}\beta 6$ / YAP1 / TAZ); DGAT herb-DDI module; QM/MM ABFE book-ending correction; spatial-transcriptomics adapter; mechanotransduction multi-tier analysis.

Tier 4 — Kinetics / SAR loops (~5 modules)

k_{off} residence-time MD setup; Chou-Talalay synergy + topical PBPK; self-improvement agent loop; Enamine REAL Tanimoto matching; Lunit / VUNO collaboration analysis.

Tier 5 — Causal / Stratified evidence (~5 modules)

Causal inference (DoWhy / EconML); papillary-reticular fibroblast stratification; skin microbiome AI; KEDD foundation-model design; federated-learning Korean-medicine-network architecture.

Tier 6 — Korean PGx + skin niche (~5 modules)

Korean PGx panel; mitochondrial PGC-1 α / TFAM; CellChat fibroblast-niche LR pairs; smartphone scar assessment; OliX siRNA collaboration.

Tier 7 — Partnership scaffolds (~5 modules)

ExoCoBio collaboration scaffold; EZH2 epigenetic adapter; NLRP3 inflammasome; Pro-Hyp triple combination; PBM LED Recover protocol.

Tier 8 — Reporting + Health Economics (~5 modules)

TRIPOD-AI checklist agent tool; SHAP / Grad-CAM explainability; 동의보감 herb miner; D+Q senolytic hair-follicle; HIRA QALY framework.

Tier 9 — Diagnostic + Specialized (~5 modules)

facial_dx integration bridge; PROTAC EMB-3 TGFB1 degrader designer; chronotherapy 자오류주; OCT bridge to facial_dx; engineered *C. acnes* probiotic; ESG / green-chemistry analyzer.

Tier 10 — Safety + MLOps + Knowledge (~5 modules)

Hallucination guard (jailbreak + medical fact citation); MLOps model registry (FDA PCCP); knowledge-graph completion; PA-EWC continual learning; multi-omics fusion.

Tier 11 — Frontier / Strategic (~6 modules)

NaFM natural-product foundation model; Bayesian adaptive trial design + n-of-1 + MFDS DTx; KIPRIS + USPTO patent prior-art; SkinAge clock; Chai-1 ensemble; JUMP-CP morphology; MFDS K-bio M&A landscape.

Total modules (Tier 0 through Tier 11): approximately 50, with substantial cross-tier dependencies. The natural-language agent interface (`src/genesis_medicine/agents/`) exposes all of these as 70 callable tools.

3. Design philosophy

3.1 Honest in silico reporting

A central commitment is **honesty about what is and is not validated**. Specific mechanisms:

- All preprints carry a “**in silico, wet-lab pending**” flag in the abstract.
- **TRIPOD-AI checklist** is run on each preprint as a self-audit.
- **Earlier incorrect results are explicitly retracted** when discovered (e.g., the earlier ABFE values on EMB-3 / Embelin under a complex-leg-only protocol [2]).
- **MMP-1 zinc handling caveat** is consistently flagged across all preprints involving MMP-1 [2,3].
- **Boltz-2 binary-classifier vs IC₅₀** distinction is consistently made.
- **자운고 narrative** correction (chemistry-based clarification of *Embelia ribes* / 자단 distinction from 자운고 / 자초) [4].

3.2 Apache-2.0 open-source

The pipeline code is open-source under Apache-2.0, which is permissive (commercial-use compatible, no GPL viral, no requirement to open-source derivative works). This contrasts with some traditional-medicine database licenses (HERB CC-BY-NC, TCMSP non-commercial) that we explicitly exclude from commercial-build profiles. The license model supports both academic publication and Recover Korean Medicine Clinic's commercial use.

3.3 3-pillar institutional integration

The pipeline is operated across three affiliated entities: - **HAN PREDICT, Inc.** (hanpredict.com) — AI healthcare platform; provides Smart Charts AI EHR + 3D facial diagnostics infrastructure. - **Genesis_Medicine Lab** — R&D division; develops and maintains the pipeline. - **Recover Korean Medicine Clinic** — clinical practice (강남, 2026-08-15 opening); provides clinical-context feedback loop.

This 3-entity integration is described in detail in our companion workflow preprint [5].

3.4 Korean-specific orientation

The pipeline is oriented to the Korean clinical, regulatory, and traditional-medicine context: - **Regulatory**: PIPA + Korean AI Basic Act + 의료법 + 화장품법 + MFDS pathways. - **Pharmacopoeial**: KP (Korean Pharmacopoeia) + KHP (Korean Herbal Pharmacopoeia) + Donguibogam digitization. - **Linguistic**: Korean clinical-context comments and natural-language tool descriptions. - **PGx**: Korean-population variant frequencies in pharmacogenomic panel.

This orientation does not preclude international application — the pipeline modules are individually general-purpose — but it does motivate specific configurations and weight choices.

4. Scaling and reproducibility

4.1 Hardware

The pipeline runs on a single-GPU configuration (1 × NVIDIA RTX 5090, 32 GB, CUDA 12.8). For reference, a typical end-to-end EMB-3-style scaffold-hopping case study (REINVENT4 100 samples + ADMET filter + Boltz-2 cofold 30 jobs + 10 ns MD + corrected ABFE 64 ns) runs in approximately 24-30 hours on this hardware.

4.2 Reproducibility

- **Conda environments** locked: `genesis-md` (MD-specific) and main `.venv` (Python 3.11).
- **Configuration as code**: `conf/*.yaml` covers all major pipeline parameters.
- **Random seeds** documented per-run.
- **Result archival**: per-run JSON outputs include configuration, version metadata, wall-clock, and key results.

4.3 License compatibility for commercial use

Two build profiles: - **Research** profile: includes HERB 2.0, TCMSP, BATMAN-TCM (non-commercial) for network-pharmacology research. - **Commercial** profile: excludes the above; relies only on permissively-licensed COCONUT 2.0 (CC0), NPASS 3.0 (CC-BY), NPAtlas (CC-BY), Dr. Duke's database (USDA public domain). Boltz-2 (MIT), REINVENT 4 (Apache-2.0), ADMET-AI (MIT), OpenMM 8 (MIT), openmmtools (MIT), RDKit (BSD-3) are all commercial-use-compatible.

This dual-profile approach permits academic and commercial use with appropriate licensing hygiene.

5. Limitations and forward path

1. **No clinical efficacy data** in any of the application studies; all are in silico hypothesis-generation.
2. **MMP-1 zinc handling** is a known limitation; ZAFF integration is planned.
3. **NaFM weights** (natural-product foundation model) are not yet integrated; placeholder fallback to Morgan fingerprint embeddings.
4. **Chai-1 integration** is interface-only; not benchmarked against Boltz-2 + Protenix consensus.
5. **JUMP-CP morphology** integration is offline-mock; AWS S3 data download (~50 GB) is the next forward step.
6. **Patent prior-art search** is offline-heuristic; KIPRIS + USPTO API integration requires API keys.

The Genesis_Medicine pipeline is a living open-source project. Contributions, issue reports, and forks are welcome under Apache-2.0.

6. Conclusions

Genesis_Medicine is an open-source, end-to-end AI pipeline for Korean traditional medicine drug discovery, integrating approximately 50+ tools across 12 functional tiers under a unified natural-language agent interface. The pipeline supports REINVENT4 generative scaffold-hopping, ADMET filtering, Boltz-2 cofold, MD validation, and corrected ABFE estimation, with explicit Korean traditional-medicine domain orientation, honest in silico reporting, and 3-pillar institutional integration with HAN PREDICT, Inc. (AI healthcare platform) and Recover Korean Medicine Clinic. The pipeline is available under Apache-2.0 at https://github.com/crazat/genesis_medicine.

We invite collaboration and contribution from the AI-drug-discovery community.

Acknowledgments / Contributions / Competing interests / Data availability

Same standard text. Code: https://github.com/crazat/genesis_medicine.

Figures

Figure 1. Genesis_Medicine 12-tier pipeline architecture overview. Approximately 50+ modules organized into 12 functional tiers from Tier 0 (Foundation: Boltz-2, REINVENT4, ADMET-AI, OpenMM, openmmtools, MACE-OFF24, RDKit) through Tier 11 (Frontier: NaFM, Bayesian trial design, patent prior-art, SkinAge clock, Chai-1, JUMP-CP, MFDS K-bio landscape). Apache-2.0 open-source.

Genesis_Medicine 12-tier pipeline (50+ modules)
Apache-2.0 open-source · github.com/crazat/genesis_medicine

Tier 0	Foundation Boltz-2, REINVENT4, ADMET-AI, OpenMM, openmmtools, MACE-OFF24, RDKit
Tier 1	Acceleration / Quality Protenix, OpenFold3, AlphaFlow-Lit, CarsDock-Cov, Boltz-ABFE, DeepRetro, AiMNet2
Tier 2	Skin-specific Adapters PanDerm, ChemCrow agent, Stratum corneum MD, Bayesian SAR, ESM3, scFoundation
Tier 3	Multi-target / Cryptic LPA1/αvβ6/YAP1/TAZ, DGAT-DDI, QM/MM ABFE, Spatial-T, mechanotransduction
Tier 4	Kinetics / SAR loops k_off MD, Chou-Talalay PBPK, self-improvement loop, Enamine REAL, Lunit/VUNO
Tier 5	Causal / Stratified DoWhy/EconML, papillary-reticular fibroblast, microbiome AI, KEDD, federated-learning
Tier 6	Korean PGx + niche Korean PGx panel, mitochondrial, CellChat, smartphone, Olix siRNA
Tier 7	Partnership scaffolds ExoCoBio, EZH2 epigenetic, NLRP3, Pro-Hyp, PBM LED
Tier 8	Reporting + HE TRIPOD-AI, SHAP/Grad-CAM, 동의보감 miner, D+Q senolytic, HIRA QALY
Tier 9	Diagnostic + Specialized facial_dx integration, PROTAC, chronotherapy, OCT, engineered probiotic, ESG
Tier 10	Safety + MLOps + KG hallucination guard, MLOps registry, KG completion, PA-EWC, multi-omics fusion
Tier 11	Frontier / Strategic NaFM, Bayesian trial + DTx, KIPRIS+USPTO patent, SkinAge, Chai-1, JUMP-CP, MFDS K-bio

Figure 1: 12-tier pipeline

References

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v0.1 draft, 2026-04-26 · ~3,200 words · CC-BY 4.0

Round 8 audit — Final 25-adapter inventory (2026-04-27)

After 4 audit rounds (Round 5 +9, Round 7 +5, Round 8 +11), the system reached **25 adapters across 14 modules**, with 14 in PRODUCTION_USE and 30+ real-data artifacts.

Module breakdown

Free-energy & ML potentials (md/, 6 adapters): ABFE corrected protocol, OpenMM ML refine, Boltz-ABFE scaffold, AToM-OpenMM scaffold, AIMNet2 scaffold, CarsiDock-Cov (covalent docker, FIRST-of-class).

Generative chemistry (generation/, 4 adapters): REINVENT 4 wrapper, FlowMol3, DecomDiff, MolDAIS BoTorch SAASBO.

Dermatology-specific (dermatology/, 4 adapters): PBK Dermal HT (NIH/NIEHS public-domain), SARA-ICE (OECD TG 497-III), Skin Fibroblast Atlas (Reynolds 2025), PanDerm (research only).

Repurposing (repurposing/, 3 adapters): TxGNN, CellAwareGNN (TxGNN successor), CMap L1000.

Conformational ensemble (ensemble/, 3 adapters): AlphaFlow, BioEmu, PocketMiner.

Foundation models (foundation/, 2 adapters): ChemBERTa-3, Tahoe-100M.

Causal inference (causal/, 1 adapter): TwoSampleMR with 7 literature-validated MR results.

Clinical workflow (clinical/, 4 adapters): IRB protocol generator, CRO quote, MedSAM-2 scaffold, IRB consent forms.

Drug-target kinetics (kinetics/, 2 adapters): τ RAMD literature-validated, SEEKR2 milestone.

Polypharmacology (polypharmacology/, 2 adapters): SwissTargetPrediction literature-validated, Dealbreaker safety panel (7 classes: hERG, CYP3A4, AR, GR, ER, PXR, AhR).

DDI (ddi/, 2 adapters): DDInter 2.0 + 17-pair curated 한약-양약.

Topical formulation (formulation/, 3 adapters): CPE-DB, HSP vehicle selection, KCID Korean cosmetic ingredient gating.

PK-PD (pkpd/, 2 adapters): htkk (US EPA public domain), Hill 4-parameter dose-response.

Protein design (protein_design/, 1 adapter): LigandMPNN scaffold for 약침 metal-aware design.

Honest categorical scoring

- Adapters with PRODUCTION_USE + literature-validated real data: 14 (56%)
- Adapters with SCAFFOLD_ONLY (graceful unavailable until external repo cloned): 11 (44%)
- Hydra YAML configs: 19/25 (76%)
- Unit tests: 29 covering Round 5+7+8 adapters (100%)

From “50 active tools” to honest 25

The earlier preprint v0.1 narrative claimed “50+ active tools.” Round 5 audit (#175 honest reframe) corrected this to “25 ACTIVE modules + 25 adapters/wrappers”. This v0.2+ count of 25 adapters is the precise audit-verified number, with 14 producing real data and 11 acting as graceful scaffolds for future external-tool integration.

Unique system contributions

1. **First DL covalent docker integration** (CarsiDock-Cov, Apache-2.0) for natural-product Michael acceptors.
2. **First Korean cosmetic regulatory adapter** (KCID + KFDA + EU CosIng cross-walk) for in silico → product launch.
3. **Single-pipeline integration** of Boltz-2 + ChEMBL $\rho=0.72$ calibration + Chai-1 ensemble + PoseBusters 149-pose validation + ABFE T4L PASS + EMB-3 cycle closed + Round 5+7+8 application data + 7 MR causal evidence + curated DDI + KCID regulatory.
4. **Honest negative result:** EMB-3 $\times$ MMP-1 ABFE +0.55 quantitatively confirms “MMP-1 minus zinc” caveat → ZAFF release-blocking.